PROJECT DESIGN PHASE

Problem – Solution Fit Template

Date	03 October 2026
Team ID	SWTID-2026-6390
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

About this template

Problem–solution fit checks whether the problem we picked is real for the users and whether our solution actually deals with it. Here it is used to show why the Institution Details table needs record-level access control in ServiceNow, and how the ACL design meets that need.

Problem – Solution Fit: Script-Controlled ACL

Controlling who can read, create, edit and delete Institution Details records, based on role and Branch value

1. CUSTOMER SEGMENTS [CS] • EEE branch users who hold the bb1 role and need to read EEE records. • Users who need to add, edit or remove records and hold bb2, bb3 or bb4. • The ServiceNow administrator who manages and tests the table.	**6. CUSTOMER CONSTRAINTS [CC]** • Access has to depend on the user's role and on the Branch value of the record. • People without the right role must not see protected records. • The administrator must keep full access. • The rules should be enforced by ServiceNow itself, not only by hiding buttons or menus.	**5. AVAILABLE SOLUTIONS [AS]** • Record-level ACLs in ServiceNow for READ, CREATE, WRITE and DELETE. • Custom roles bb1, bb2, bb3 and bb4. • An advanced READ script that uses gs.hasRole(). • A data condition on the READ ACL: Branch = EEE.
2. JOBS-TO-BE-DONE / PROBLEMS [JP] • Keep users from opening Institution Details records outside the branch they are allowed to see. • Let EEE users work with EEE records. • Give create, edit and delete rights separately instead of one combined permission. • Not depend on the user interface alone to hide data.	**9. PROBLEM ROOT CAUSE [RC]** • A new custom table does not limit records by branch on its own. • A role says what a user may do, but it does not check the Branch value of each record. • Without ACL rules, records could be seen or changed by users who should not have access.	**7. BEHAVIOR [BE]** • A bb1 user opens the Institution Details list and gets only EEE records. • Creating needs bb2, editing needs bb3 and deleting needs bb4. • If a user lacks the role for an operation, that operation is denied. • The administrator can use all records whatever the branch is.
3. TRIGGERS [TR] • A user tries to open, add, edit or delete an Institution Details record. • ServiceNow checks the matching ACL before it lets the action happen.	**4. CUSTOMER EFFECT / AFTER [EM]** • Each user gets only the access that their role needs. • Records of other branches stay protected from viewing and changes. • Users and administrators can confirm the result by testing.	**8. CHANNELS & BEHAVIORS [CH]** • Application Navigator and the Institution Details list in ServiceNow. • Record forms and list actions such as New, Edit and Delete. • Impersonation to test each role combination. • ACLs are checked on the server side, so the rules apply to every way of reaching the table.

10. YOUR SOLUTION [SL]

• Create the custom table u_institution_details (Institution Details) with a Branch choice field containing ECE, EEE and CSE.
• Create the roles bb1, bb2, bb3 and bb4 and give them to users according to the access they need.
• READ ACL: requires bb1, applies the condition Branch = EEE, and uses an advanced script that allows the administrator and bb1 users.
• CREATE ACL requires bb2, WRITE ACL requires bb3, and DELETE ACL requires bb4.
• Impersonate different users and compare what they can do with what the administrator can do, to confirm the rules work.